The $(\omega_{\mathrm{qAFM}},\, \omega_{E_{12}})$ Cross-Peak in $TmFeO_3$: Diagnosis of the Route A Campaign and a Modeling Roadmap

ClassicalSpin_Cpp project notes

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Abstract

We set out to confirm or deny a nonlinear 2DCS cross-peak at $(\omega_{\mathrm{qAFM}},\, \omega_{E_{12}}) \approx (0.906,\, 0.500)$ THz produced by the magnetoelastic / bilinear coupling χ_{2x} between the Fe qAFM magnon and the Tm $E_1 \to E_2$ crystal-field transition. After rewriting the analysis to use the correct observable channel ($\Lambda^1_G = 4\,\overline{\lambda^1_{\mathrm{local}}}$), running the $\chi_{2x} = 0$ baseline (identically zero nonlinear signal), and fixing a units error in the τ-domain analysis, we conclude: **at the campaign value $\chi_{2x} = 0.75$ meV there is no detectable cross-peak at the bare $E_{12} = 0.500$ THz.** The reason is physical, not numerical: the static mean field generated by χ_{2x} is large compared to the bare crystal-field splitting and *renormalizes the Tm transition* down to ≈ 0.074–0.147 THz, so the spectral weight that should land at E_{12} has moved. This document records the full chain of evidence and then lays out three concrete modeling routes to recover a clean, interpretable cross-peak.

Contents

1 Physical setup and notation

$TmFeO_3$ (space group *Pbnm*, No. 62) in the Γ_2 phase carries two coupled magnetic sectors:

- **Fe** (Wyckoff $4b$, $S = 5/2$, SU(2), `spin_dim`=3): hosts the magnon branches $\omega_{\mathrm{qFM}} \approx 0.378$ THz and $\omega_{\mathrm{qAFM}} \approx 0.906$ THz.
- **Tm** (Wyckoff $4c$, non-Kramers qutrit, SU(3), `spin_dim`=8): hosts the crystal-field (CEF) transitions $E_{12} \approx 0.500$ THz ($e_1 = 2.068$ meV) and $E_{13} \approx 1.200$ THz ($e_2 = 4.963$ meV).

The mixed state vector has `state_dim`=44: indices 0–11 are the 4 Fe sites × 3 components; indices 12–43 are the 4 Tm sites × 8 Gell-Mann components $\lambda^1 \ldots \lambda^8$.

Units (critical). The integrator works in natural units $\hbar = 1$ with energies in meV:

- Time is measured in $\hbar$/meV; one code time unit = 0.6582 ps.
- A frequency returned by `numpy.fft.fftfreq(N, dt)` with `dt` in code units is therefore in **meV**.
- Convert to THz with $f_{\mathrm{THz}} = f_{\mathrm{meV}} \times 0.241799$.

The target cross-peak in code units is $\omega_{\mathrm{qAFM}} = 3.748$ meV, $\omega_{E_{12}} = 2.068$ meV.

2 Route A: the intended mechanism

The proposed ("Route A") mechanism for the cross-peak is a two-step nonlinear cascade driven by the bilinear Fe–Tm coupling χ_{2x}:

1. Pump 1 (THz, along the appropriate polarization) launches the Fe **qAFM** magnon. This modulates the Fe order parameter at ω_{qAFM}.
2. Through χ_{2x}, the oscillating Fe moment drives the Tm σ_F even channel (λ^2, sublattice-uniform), which in turn shifts the Tm λ^1 coherence at the CEF frequency E_{12}.
3. In a 2DCS scan over inter-pulse delay τ, the qAFM phase imprinted during step 1 should appear as modulation of the E_{12} response, producing a peak at $(\omega_{\mathrm{qAFM}}, \omega_{E_{12}})$.

Observable channel (confirmed). The nonzero Tm dipole channel in the Γ_2 ground state is the global G-type λ^1, which equals four times the site-averaged local-frame λ^1:

$$\Lambda^1_G = \sum_{k=0}^{3} \lambda^1_{\mathrm{local}}(k) = 4\,\overline{\lambda^1_{\mathrm{local}}} = 4\,\big[\texttt{M\_local\_SU3}\big]_{:,0}. \tag{1}$$

Empirically (at $\tau = -100$, $t = t_0$, $\chi_{2x} = 0.75$): $\lambda^1_{\mathrm{local}}(k) = [-9.664, -9.664, -9.699, -9.699] \times 10^{-6}$ (uniform σ_F sign pattern), giving $\Lambda^1_G = -3.87 \times 10^{-5}$, while $\Lambda^1_C \approx 6.9 \times 10^{-8} \approx 0$.

> [WARN] **Naming bug in older notes.** The earlier "Λ^1_C" that was claimed to carry the cross-peak is a mislabel. The nonzero channel is $\Lambda^1_G = 4\,\overline{\lambda^1_{\mathrm{local}}}$; the true Λ^1_C is ≈ 0 by symmetry. The global-frame reader channel `M_global_SU3`$_{:,0} \approx 0$ averages to zero and yields only a noise-level spectrum — this is why the precomputed `M_NL_SU3_FF_`λ`1.npy` files are useless for this search.

3 What we did and what we found

3.1 Baseline: $\chi_{2x} = 0$ gives identically zero nonlinear signal

The run `v3_grid_chi2x/v3_grid_chix_0_W_0` produces NL[λ^1] = NL[λ^2] = 0 exactly (rms = 0.000). This confirms (a) the nonlinear subtraction NL = $M_{01} - M_0 - M_1$ is clean, and (b) all `v3_grid_chi2x` campaigns have $\chi_{2z} = 0$ (no residual even coupling).

3.2 $\chi_{2x} = 0.75$, $W = 0$: the 2D spectrum

Using `util/crosspeak_routeA.py` on the 4.2 GB `v3_grid_chix_0p75_W_0` file, with correctly calibrated **linear** THz axes (§3.6) and Hann window (default after the fix):

Position (ω_τ, ω_t) [THz, linear]	Channel	Amplitude	S/N
$(0,\ f_2 = 0.147)$ DC-in-τ stripe	λ^1_{loc}	2.39×10^{-3}	23.4
$(\text{qAFM} = 0.906,\ f_2 = 0.147)$ dressed cross-peak	λ^1_{loc}	$\sim 10^{-2}$(a)	∼4–7(a)
$(\text{qAFM},\ E_{12} = 0.500)$ bare CEF target	λ^1_{loc}	9.96×10^{-7}	< 0.1
Background rms	λ^1_{loc}	1.02×10^{-4}	—
$(0,\ f_1 = 0.074)$ global-frame DC	λ^1_{FF} (reader)	—	137
$(\text{qAFM},\ f_1 = 0.074)$	λ^1_{FF}	—	0.14
$(\text{qAFM},\ E_{12} = 0.500)$	λ^1_{FF}	—	< 0.01

(a) Hann or no-apod window required; Gaussian window $\gamma = 0.03$ destroys this peak (see §3.6).

[FAIL] The target $(\omega_{\text{qAFM}}, \omega_{E_{12}})$ sits *at* the noise floor (S/N = 1.1). The only genuine feature is the Fe magnon diagonal at (qAFM, qAFM) (S/N = 4.3). The global maximum is a ridge that is *flat in* ω_τ at $\omega_t = \omega_{\text{qAFM}}$ — the signature of a signal that is constant in τ ("DC-in-τ"), whose FFT leaks across all ω_τ.

3.3 The key discovery: χ_{2x} renormalizes the Tm transition

Inspecting the nonlinear λ^1 $(= \lambda^1_{\text{local}})$ response in the time domain at fixed τ, the dominant oscillation is *not* at $E_{12} = 0.500$ THz but at

$$f_1 \approx 0.074 \text{ THz } (0.306 \text{ meV}), \qquad f_2 \approx 0.147 \text{ THz } (0.608 \text{ meV}), \qquad f_2/f_1 \approx 1.99. \tag{2}$$

The amplitudes at f_1, f_2 exceed those at E_{12} and qAFM by 250–500×. The origin is a large static mean field: with $\chi_{2x} = 0.75$ meV and $\langle S \rangle \sim$ several,

$$\chi_{2x} \langle S^2 \rangle \ \sim\ 6 \text{ meV } \gg\ e_1 = 2.07 \text{ meV} = \hbar\omega_{E_{12}}, \tag{3}$$

so the Tm two-level subspace is strongly dressed and its effective splitting collapses to $\approx f_1$. Consequently the spectral weight that "should" appear at E_{12} has physically moved to f_1, and the 2D FFT shows *nothing* at the bare E_{12}.

3.4 The equilibrium itself is not stationary

The unpumped reference M_0 already oscillates: λ^1 of the reference trajectory rings persistently at f_1, f_2 with no decay (early/late amplitude ratio ≈ 1.07 over the full -65.8 to $+65.8$ ps window), while the populations (λ^3, λ^8) stay constant. This means the Tm qutrit was **initialized in the bare CEF eigenbasis, not in the χ_{2x}-dressed eigenbasis**; the mismatch launches a coherent ringing at the dressed frequency. This ringing contaminates the nonlinear subtraction and is the source of the strong f_1, f_2 lines.

3.5 Units error that previously masked the picture

An earlier τ-FFT used `stride`=4 (effective $d\tau_{\text{eff}} = 0.2$ code units), giving Nyquist = 2.5 meV = 0.605 THz. Because $\omega_{\text{qAFM}} = 3.748$ meV is *above* this, the qAFM modulation was **aliased**, and a "dominant 0.609 THz tau-oscillation" was reported that was actually 0.608 meV = 0.147 THz = f_2. With `stride`=1 and correct meV→THz conversion, the genuine complex amplitude at E_{12} does carry an $\omega_\tau = \omega_{\text{qAFM}}$ Fourier component at $\sim 8\%$ of its DC value — real, but spread so broadly in the 2D map that it never rises above the noise floor.

3.6 Analysis bugs discovered and fixed

Three analysis bugs were found and corrected in `util/crosspeak_routeA.py` after running the campaign. Understanding them is essential for correctly interpreting all previous S/N numbers.

Bug 1: Angular vs. linear THz (critical). The `do_fft2()` function saved omega axes as $\omega_{\mathrm{angular}} = f_{\mathrm{code}} \times 2\pi \times 0.241799$ (rad/ps), but the search constants were defined in *linear* THz ($\nu_{\mathrm{qAFM}} = 0.906$, $\nu_{E_{12}} = 0.500$ THz). The search was therefore looking at

$$\nu_{\mathrm{wrong}} = \frac{\omega_{\mathrm{angular}}}{2\pi} \approx 0.906/(2\pi) \approx 0.144 \text{ THz} \approx f_2,$$

i.e. *at the dressed Tm frequency*, not at qAFM! All previous S/N values at the “qAFM position” were actually measuring the dressed-Tm response. Fix: remove the 2π factor; save in linear THz.

Bug 2: Flip/index mismatch in ω_τ. The spectrum rows are flipped (`spec[::-1,:]`, rephasing convention), but the saved `omega_tau` array was the *unflipped* DFT frequency grid. Row i of the saved spectrum corresponded to $\omega_\tau[n_\tau - 1 - i]$, not $\omega_\tau[i]$, causing the positive-quadrant search to look in the wrong direction. Fix: save `otau = otau_unflipped[::-1]`, so index i maps consistently to the i-th row of the flipped spectrum.

Bug 3: Gaussian apodization destroys the qAFM cross-peak. The default Gaussian window has $\sigma_\tau = \tau_{\mathrm{range}}/k \approx 24.9$ ps. The product

$$\sigma_\tau \cdot \omega_{\mathrm{qAFM}}^{\mathrm{code}} = 37.76 \text{ (code)} \times 3.748 \text{ (code}^{-1}) \approx 141 \gg 1$$

places ω_{qAFM} far in the tail of the Gaussian spectral envelope. The DFT at the ω_{qAFM} bin is dominated by the DC sidelobe, not the genuine qAFM modulation (integration-by-parts estimate: $\sim 1/(\omega_{\mathrm{qAFM}}\, d\tau) \approx 5\%$ of DC versus the $\approx 1\%$ true cross-peak). Without apodization or with a Hann window (which has zero-mean sidelobes in each basis direction), the cross-peak at (qAFM, $f_2 = 0.147$) THz emerges with S/N$\approx$ 4–7. Fix: default apodization changed from Gaussian(γ=0.03) to Hann; `-window {hann,gaussian,none}` option added.

> [WARN] The S/N table in the previous version of this document (with Gaussian apodization and angular-THz axes) is **incorrect**. The corrected table is in §3.2 above. Old files in `build/workflow/routeA_analysis/omega_*.npy` store angular THz; divide by 2π before use.

3.7 Updated physical picture at $\chi_{2x} = 0.75$ meV

With all bugs fixed:

- The dominant NL feature is the DC-in-τ stripe at $\omega_t = f_2 = 0.147$ THz: the dressed Tm oscillation is excited only at $\tau = 0$ (coincident pumps) and thus has no τ modulation under Gaussian apodization.
- With Hann or no-apodization window: a genuine cross-peak is detected at $(\omega_\tau, \omega_t) = (0.906, 0.147)$ THz (i.e. qAFM vs. dressed f_2) with S/N $\approx$ 4–7. This confirms the Route A mechanism is active.
- No peak at the bare CEF target $(0.906, 0.500)$ THz in any windowing scheme, consistent with strong coupling: $\chi_{2x}\langle S^2\rangle \approx 6$ meV $\gg e_1 = 2.07$ meV.

4 Diagnosis summary

Bottom line. At $\chi_{2x} = 0.75$ meV the Route A mechanism *is confirmed*: a cross-peak exists at $(\omega_{\text{qAFM}},\ f_2) = (0.906,\ 0.147)$ THz at S/N $\approx$ 4–7 (Hann/no-apod window). However, this is the *dressed* frequency — the strong mean field ($\chi_{2x}\langle S^2\rangle \approx 6$ meV $\gg e_1 = 2.07$ meV) renormalizes the Tm transition from 0.500 THz down to $f_2 = 0.147$ THz. A Gaussian apodization window with $\gamma = 0.03$ (equivalent to $\sigma_\tau \cdot \omega_{\text{qAFM}} \approx 141$) destroys the cross-peak via DC sidelobe leakage; the fix is to use a Hann window (new default in the script). The bare-CEF cross-peak at (0.906, 0.500) THz is absent in all windowing schemes — to see it we must work in the perturbative regime ($\chi_{2x} \ll e_1/(\langle S^2\rangle) \approx 0.35$ meV).

5 Modeling roadmap: how to recover the cross-peak

The goal is a 2DCS map in which a peak appears at a *predictable* (ω_τ, ω_t) that we can attribute to qAFM $\to E_1E_2$ energy transfer through χ_{2x}. There are three complementary routes, in order of increasing effort.

5.1 Route 1 — Weak-coupling / perturbative regime (fastest)

Reduce χ_{2x} so that the dressing is perturbative, $\chi_{2x}\langle S^2\rangle \ll e_1$. Then the Tm transition stays at the bare E_{12} and the cross-peak amplitude is, to leading order,

$$I_{\text{cross}} \;\propto\; \chi_{2x}^2\, \frac{\partial \langle S_{\text{qAFM}}\rangle}{\partial E_1}\Big|_{\text{pump1}} \;\times\; \mu_{12}^2, \tag{4}$$

i.e. it grows as χ_{2x}^2 while the frequency stays pinned at E_{12}.

- **Action.** Run a small grid $\chi_{2x} \in \{0.02,\ 0.05,\ 0.10,\ 0.20\}$ meV at $W = 0$.
- **Expectation.** The peak should sit at $(0.906, 0.500)$ and its intensity should scale as χ_{2x}^2. Confirming the quadratic scaling is the cleanest possible proof that the peak is the Route A cascade and not an artifact.
- **Sampling.** Keep $d\tau$ fine enough that $\omega_{\text{qAFM}} = 3.748$ meV is below Nyquist: $d\tau < 1/(2 \times 3.748) = 0.133$ code units. The campaign value $d\tau = 0.05$ is fine.

5.2 Route 2 — Dress the reference, then linearize (most physical)

Keep $\chi_{2x} = 0.75$ but *remove the spurious ringing* so the spectrum is clean at the (renormalized) Tm frequency.

1. **Relax to the dressed ground state.** Before launching pumps, evolve (or minimize) the coupled Fe+Tm system to its true equilibrium so that the reference $M_0(t)$ is stationary. Practically: add a long damped pre-roll, or solve the self-consistent mean-field fixed point $\partial\mathcal{H}/\partial\lambda = 0$ for the Tm qutrit in the static Fe field, and initialize the qutrit in *that* eigenbasis.
2. **Recompute the dressed transition** $\tilde{\omega}_{12}$ from the Hessian of $\mathcal{H}$ at the fixed point (this is the 0.074–0.147 THz line). The cross-peak will then appear at $(\omega_{\text{qAFM}},\ \tilde{\omega}_{12})$, which is a legitimate physical observable — just at a renormalized frequency.
3. **Verify** by overlaying the linear (1D) Tm spectrum: the single pump response must show a peak exactly at $\tilde{\omega}_{12}$.

This route documents the renormalization rather than fighting it, and is the correct thing to report if the experimental χ_{2x} really is $\sim$ meV.

5.3 Route 3 — χ_{2z}-mediated even channel (alternative cascade)

All current campaigns have $\chi_{2z} = 0$. The physical value is $\chi_{2z} \approx 0.6$. The χ_{2z} coupling drives the Tm *even* (λ^2, G-type) channel directly and can produce a cross-peak with a different, possibly cleaner, selection rule.

- **Action.** Generate a new campaign with $\chi_{2z} = 0.6$, $\chi_{2x} = 0$ (isolate the even cascade), then $\chi_{2x} = \chi_{2z}$ together.
- **Expectation.** The even channel couples $\langle S^z_{\mathrm{qAFM}}\rangle^2$ to the λ^2 population, giving a rectified (DC + $2\omega_{\mathrm{qAFM}}$) drive of the Tm levels; the cross-peak structure differs from Route A and should be compared.

5.4 Analysis-side fixes (apply to all routes)

- **Omega axes: always use linear THz.** Multiply `numpy.fft.fftfreq` output by `MEV_TO_THZ` (not $2\pi\times$`MEV_TO_THZ`). Label axes $\nu = \omega/(2\pi)$ (THz, cycles/ps). **[OK]** Fixed in `do_fft2()` (see §3.6).
- **Flip/index consistency.** After the rephasing flip `spec[::-1,:]`, save the *reversed* `omega_tau` so that row i maps to `omega_tau[i]`. **[OK]** Fixed: `otau = otau_unflipped[::-1]`.
- **Use Hann (or no) apodization in τ.** Gaussian($\gamma = 0.03$) with $\sigma_\tau \cdot \omega_{\mathrm{qAFM}} \approx 141$ kills the cross-peak. Hann window gives S/N ≈ 4 at the qAFM cross-peak. **[OK]** Fixed: new default `-window hann` in the script.
- **Remove DC-in-τ before the 2D FFT.** Subtract the τ-mean at each t: $\widetilde{\mathrm{NL}}(\tau, t) = \mathrm{NL}(\tau, t) - \overline{\mathrm{NL}}(\cdot, t)$. This kills the flat ridge that dominates the raw map.
- **Use `stride`=1 in τ** (or verify $d\tau < 0.133$) so ω_{qAFM} is not aliased.
- **Report S/N against the local rms**, and treat anything below $\sim 3\sigma$ as absent.

6 Concrete next experiment

The recommended first step is **Route 1**, because it is decisive and cheap:

1. Run the $\chi_{2x} \in \{0.02, 0.05, 0.10, 0.20\}$ grid, $W = 0$, $\chi_{2z} = 0$, $d\tau = 0.05$, same t-grid as the current campaign.
2. For each, run `crosspeak_routeA.py` *with the new DC-in-τ subtraction.*
3. Plot $I(0.906, 0.500)$ vs. χ_{2x} on log–log axes. A slope of 2 confirms the Route A χ_{2x}^2 cascade and a peak pinned at the bare E_{12}.

If the peak frequency *drifts downward* with increasing χ_{2x} across this grid, that directly visualizes the renormalization and motivates Route 2 for the strong-coupling endpoint.

A File and data inventory

- Analysis script: `util/crosspeak_routeA.py` (reads `M_local_SU3`, computes 2D NL FFT). Updated: linear THz axes, Hann default window, flip fix. Options: `-window {hann,gaussian,none}`.
- Visualization: `util/visualize_2dcs_chi2x075.py` (loads precomputed npy, produces 3-panel figures with linear THz axes).
- $\chi_{2x} = 0.75$, $W = 0$ data: `build/workflow/v3_grid_chi2x075_wline/v3_grid_chix_0p75_W_0/sample_0/pump_probe_spectroscopy.h5` (4.2 GB).
- Precomputed reader npy: `M_NL_SU3_FF_λ*.npy` (global-frame, angular THz; divide by 2π).
- Baseline $\chi_{2x} = 0$: `build/workflow/v3_grid_chi2x/v3_grid_chix_0_W_0/`.
- Stored routeA spectra: `build/workflow/routeA_analysis/`
  `omega_*.npy`: **angular THz (old)** — divide by 2π or re-run script.
  `spec_lambda*.npy`: spectral amplitudes (valid; omega stored separately).
  `fig1_2d_spectra.pdf/.png`, `fig2_diagnostics.pdf/.png`, `fig3_full_range.pdf/.png`, `fig4_no_apod_hann.png`: diagnostic figures.

B Lambda-index key (`M_local_SU3`)

Index	Generator	Meaning / equilibrium value at $\chi_{2x} = 0.75$
0	λ^1	E_{12} x-coherence; -5.24×10^{-3}
1	λ^2	E_{12} y-coherence; -1.35×10^{-2}; DC-dominated NL
2	λ^3	diagonal population; 0.415
3–6	$\lambda^{4..7}$	E_{13}/E_{23} off-diagonal; 0 when $\chi_{2z} = 0$
7	λ^8	total population; 0.910 (constant)